

Loan Recovery Voice Agent

VICIdial Integration Guide

For FastAPI + WebSocket + STT/TTS + Gemini recovery agent

Prepared for implementation and server deployment

Recommended production path

Use VICIdial/Asterisk for dialing, campaign control, call records, and dispositions. Use your AI agent as the conversation brain. For real customer calls, add a server-side telephony bridge between Asterisk media and your agent. Do not rely on the browser microphone path for production customer audio.

1. Scope and integration decision

This guide explains how to connect the current loan recovery voice agent to VICIdial. The agent side is assumed to be the FastAPI/WebSocket stack you have been running locally: browser call page, WebSocket voice session, Sarvam STT/TTS, Gemini dynamic response generation, and customer/lead data supplied either from a form or from data.json.

What VICIdial does vs. what the agent does

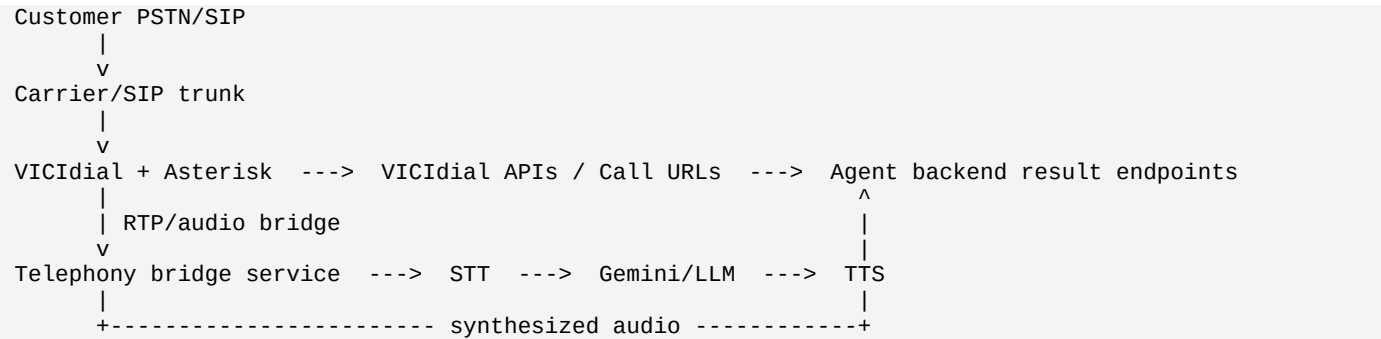
Layer	Responsibility	Implementation owner
VICIdial	Campaigns, lead loading, dialing, call routing, call records, agent/disposition workflow.	Dialer admin / telephony team
Asterisk under VICIdial	SIP/RTP media, dialplan execution, IVR/AGI/ExternalIVR integration.	Telephony engineer
AI recovery agent	Conversation state, LLM response generation, PTP decisioning, STT/TTS, notes, result payloads.	Python backend team

Choose one of three integration modes

Mode	Purpose	What connects	When to use
Mode A - Browser-assisted	Fastest proof of concept	VICIdial opens your /call page with lead data; human/browser still owns mic and speaker.	Demo, QA, manual collection desk assistance.
Mode B - Server-side voice bot	Production automation	Asterisk audio is bridged to the Python agent; TTS audio is sent back to the call.	Automated outbound/inbound recovery calls.
Mode C - Hybrid agent assist	Human plus AI	VICIdial agent screen launches AI panel; human can monitor, interrupt, or take over.	Compliance-sensitive or semi-automated workflows.

2. Target architecture

For production, treat VICIdial as the dialer and call-control layer, and treat the AI service as a media/conversation service. The browser WebSocket endpoint is useful for testing, but production customer audio should not depend on a browser tab.



Key control paths

- Lead/control data path: VICIdial sends lead_id, phone_number, campaign_id, user, caller_code, and other fields to the agent backend through Start URL, Web Form URL, API calls, or a custom dialplan argument.

- Audio path: Asterisk/VICIdial call media is connected to the agent through AGI/EAGI, ExternalIVR, ARI External Media, AudioSocket, or a SIP bridge. This is separate from VICIdial web APIs.
- Result path: the agent posts PTP outcome, refusal, escalation, wrong-person, or call-ended result back to your backend and optionally to VICIdial through Agent API or Non-Agent API.

3. Mode A - Browser-assisted VICIdial launch

This is the fastest integration because it does not require a custom Asterisk media bridge. Use it to load the AI call page with VICIdial lead data and let a browser session run the AI voice interface. It is not a full bot-to-customer telephony integration.

Steps

1. Deploy the FastAPI app on a reachable HTTPS URL, for example <https://agent.example.com>.
2. Expose the call UI at /call and the WebSocket endpoint at /ws/call.
3. Add query-parameter support in call.html so it can prefill first_name, phone_number, loan_id, due_amount, and lead_id from the URL.
4. In VICIdial campaign or in-group settings, configure Web Form or Start Call URL to open or call your agent URL with VICIdial variables.
5. Test with one campaign/list first. Confirm the correct lead is loaded and the agent state starts at identity confirmation.

Example Web Form / Start URL

Use your actual domain and only the fields you need. In VICIdial URL fields, use the VICIdial variable wrapper format supported by your installation.

```
VARhttps://agent.example.com/call?
lead_id=--A--lead_id--B--&phone=--A--phone_number--B--&loan_id=--A--vendor_lead_code--B--&
first_name=--A--first_name--B--&campaign=--A--campaign--B--&user=--A--user--B--
```

Important limitation: this mode launches the UI and passes data, but it does not automatically connect the customer RTP stream into your WebSocket agent. For actual autonomous calls, use Mode B.

4. Mode B - Production server-side voice bot

This is the recommended architecture when the AI agent must speak directly to customers dialed by VICIdial. You need a telephony bridge that takes audio from Asterisk, sends it to STT, sends transcript to the agent, receives the bot text, synthesizes TTS, and plays it back to the same call.

Bridge options

Option	How it works	Pros	Notes
AGI/EAGI	Asterisk executes a script from dialplan. EAGI can read call audio from file descriptor 3.	Common, simple, works with older Asterisk.	Good first production bridge if your Asterisk version is older.
ExternalIVR	Asterisk connects to a local app or TCP socket process that receives events and sends commands.	Designed for external IVR applications.	Good fit if you want a long-running socket server.
ARI External Media / AudioSocket	Asterisk sends media to an external application over RTP/WebSocket-like media plumbing depending on version/module.	Cleaner media architecture, scalable.	Requires validating Asterisk module/version compatibility in your VICIdial build.

SIP bot endpoint	The bot acts like a SIP endpoint/phone and receives transferred calls.	Easy to reason about call routing.	You still need media handling, STT/TTS, and call state inside the bot service.
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Minimal Asterisk dialplan concept

Do not paste this blindly into production; adapt context names to your VICIdial/Asterisk installation. The intent is to show the media handoff point.

```
[loan-bot]
exten => 8600,1,NoOp(Loan Recovery AI Bot)
same => n,Answer()
same => n,Set(LEAD_ID=${CALLERID(name)})
same => n,AGI(loan_recovery_agi.py,${LEAD_ID},${CALLERID(num)},{UNIQUEID})
same => n,Hangup()

; ExternalIVR variant
exten => 8601,1,NoOp(Loan Recovery ExternalIVR Bot)
same => n,Answer()
same => n,ExternalIVR(ivr://127.0.0.1:2949(${CALLERID(num)},{UNIQUEID}),n)
same => n,Hangup()
```

Bridge service responsibilities

- Read 8 kHz telephony audio from Asterisk and resample to the STT input rate, commonly 16 kHz PCM16.
- Apply VAD/end-of-turn detection to avoid sending silence to STT.
- Call your agent session with text, not browser audio. The existing RecoveryAgentSession.handle_user_text(text) pattern is the right core interface.
- Synthesize the bot_text using TTS, resample/output to the format Asterisk expects, and play it to the channel.
- Capture per-turn latency: STT milliseconds, LLM milliseconds, TTS milliseconds, total turn time.
- Post the final outcome to your result endpoint and/or update VICIdial disposition.

5. Conversation workflow mapping

Keep the Python workflow deterministic and let Gemini generate only the spoken wording. This preserves compliance and prevents the LLM from creating invalid PTP dates or skipping required identity/payment steps.

Workflow stage	Trigger	Agent action	VICIdial result/status
identity	Call starts or customer response unclear	Confirm correct borrower. If wrong person, close politely.	WRONG_PERSON or continue
initial	Identity confirmed	State EMI due and ask for today payment.	Continue
refusal -> PTP offer	Customer says cannot pay today	Do not ask today-payment again. Directly offer PTP using policy max days.	PTP_PENDING
negotiation	Customer accepts or negotiates PTP	Create PTP only after explicit confirmation.	PTP_CREATED
no_ptp	Bucket exhausted or PTP rejected repeatedly	Push partial/same-day payment and escalate after max refusals.	ESCALATED / NO_PTP
closed	Committed, PTP created, wrong person, customer ended, escalation	End call and save transcript/outcome.	Final disposition

6. VICIdial API and URL integration

Use VICIdial APIs for control data and dispositions, not for live audio. Live audio needs a telephony bridge. The two useful API families are the Agent API and Non-Agent API.

Inbound data endpoint on your agent backend

```
POST /vicidial/start
{
  "lead_id": "12345",
  "phone_number": "9991110001",
  "first_name": "Rahul Patel",
  "loan_id": "701",
  "due_amount": 2500,
  "campaign_id": "LOAN_RECOVERY",
  "caller_code": "...",
  "uniqueid": "..."
}
```

Final result endpoint on your agent backend

```
POST /vicidial/result
{
  "lead_id": "12345",
  "call_id": "abc123",
  "outcome": "PTP_CREATED",
  "ptp_days": 3,
  "ptp_date": "2026-06-07",
  "due_amount": 2500,
  "summary": "Customer confirmed 3-day PTP.",
  "latency": {"stt_ms": 840, "llm_ms": 620, "tts_ms": 1250, "total_ms": 2710}
}
```

Example VICIdial disposition update calls

The exact function names and parameters must be checked against your VICIdial version and API user permissions. Keep these calls server-side; do not expose API credentials in the browser.

```
# Non-Agent API style lead update concept
GET https://VICIDIAL_HOST/vicidial/non_agent_api.php?
source=loanbot&user=APIUSER&pass=APIPASS&function=update_lead&lead_id=12345&status=PTP&comments=PTP
%203%20days%20to%202026-06-07

# Agent API style disposition concept when controlling an active agent session
GET https://VICIDIAL_HOST/agg/api.php?
source=loanbot&user=APIUSER&pass=APIPASS&agent_user=BOT001&function=external_status&value=PTP
```

7. Required configuration

Agent server .env

```
# Agent core
AGENT_PUBLIC_URL=https://agent.example.com
AGENT_USE_DYNAMIC_LLM=1
AGENT_STRICT_DYNAMIC_LLM=1
AGENT_DEBUG_LOGS=1

# Gemini
GEMINI_API_KEY=replace_with_new_key
GEMINI_MODEL=gemini-3.5-flash
LLM_TEMPERATURE=0.35
LLM_RETRY_COUNT=1
LLM_RETRY_DELAY=0.25
```

```
# Sarvam
SARVAM_API_KEY=replace_with_sarvam_key

# VICIdial
VICIDIAL_BASE_URL=https://vicidial.example.com
VICIDIAL_API_USER=loanbot_api
VICIDIAL_API_PASS=replace_with_secret
VICIDIAL_SOURCE=loanbot
VICIDIAL_DEFAULT_CAMPAIGN=LOAN_RECOVERY
```

Network/security checklist

- Use HTTPS for all browser/API endpoints. If VICIdial and the agent are on private infrastructure, still use TLS or a VPN.
- IP-allowlist VICIdial to call /vicidial/start and /vicidial/result.
- Create a dedicated VICIdial API user with only required API functions and user-group scope.
- Do not pass Gemini, Sarvam, or VICIdial API secrets through the browser.
- Treat loan data, phone numbers, recordings, and transcripts as sensitive PII. Set retention and access policies.
- Avoid writing full API URLs with passwords to normal application logs.

8. Deployment sequence

1. Deploy the current FastAPI agent on the same LAN/VPN as VICIdial first. Confirm /health responds.
2. Verify Gemini and Sarvam keys: backend logs should show llm_ready, not llm_disabled.
3. Enable latency logs and run text-only tests using /test/text before connecting calls.
4. Configure Mode A browser launch first. Confirm lead data loads correctly from VICIdial URL variables.
5. Build the server-side media bridge and test with a single Asterisk extension, outside production campaigns.
6. Route one internal test DID or campaign to the bot extension. Confirm audio, end-of-turn detection, TTS playback, hangup handling, and outcome saving.
7. Add result/disposition update to VICIdial. Test PTP_CREATED, COMMITTED, WRONG_PERSON, CUSTOMER_ENDED, and ESCALATED.
8. Pilot with one campaign, low dial ratio, and full call recording/transcript review.
9. Scale only after average latency, one-way audio rate, and disposition accuracy are acceptable.

9. Latency logging and optimization

For every customer turn, print and store separate timings. This makes it obvious whether delay is coming from STT, the LLM, TTS, or audio/VAD buffering.

```
[LATENCY] call_id=abc123 turn=4 stt_ms=820 llm_ms=610 tts_ms=1240 agent_ms=630 total_ms=2850
```

What to optimize first

- VAD/end-of-turn: excessive trailing silence makes the bot feel slow even if STT/LLM/TTS are fast.
- Prompt size: keep each LLM prompt compact; send state and guidance, not huge transcripts.
- LLM retries: in live calls, retry_count=1 is usually better than waiting multiple seconds.
- TTS: keep bot responses to one short sentence whenever possible.
- Bridge location: keep the telephony bridge close to Asterisk and the STT/TTS providers to reduce network jitter.
- Streaming: use streaming STT/TTS if your provider and bridge support it.

10. Test cases before production

Test	Customer phrase	Expected bot behavior	Expected outcome
Identity match	हाँ, मैं राहुल पटेल बोल रहा हूँ।	Moves to due EMI prompt.	Continue
Wrong person	गलत नंबर है।	Apologizes and ends.	WRONG_PERSON
Cannot pay	नहीं, आज payment नहीं कर पाऊंगा।	Does not ask today-payment again; directly offers PTP.	PTP_PENDING
PTP accepted	हाँ, 3 दिन का PTP set कर दीजिए।	Creates PTP and closes.	PTP_CREATED
PTP exhausted	मुझे time चाहिए।	Says no extension available and asks for payment/partial.	NO_PTP
Question first	Fine कितना लगेगा?	Answers briefly, then returns to payment/PTP flow.	Continue
Customer ends	Call बंद करो।	Thanks and closes.	CUSTOMER_ENDED
Emergency	Medical emergency है।	Escalates politely.	ESCALATED

11. Troubleshooting

Symptom	Likely cause	Fix
llm_disabled	GEMINI_API_KEY missing or google-generativeai package unavailable.	Set .env in the server working directory, install package, restart Uvicorn.
403 leaked key	Gemini key has been blocked by provider.	Create a new key and remove the leaked key from .env, logs, and git history.
Start URL does not fire	Manual dial or wrong campaign/in-group field.	Use Web Form for manual UI launch; Start URL is for applicable auto/inbound call flows.
One-way audio	NAT/RTP/firewall issue between carrier, Asterisk, and bridge.	Check SIP signaling, RTP port range, extenip/localnet, and packet captures.
Bot repeats payment question after refusal	Prompt allowed a binary pay-or-PTP branch after clear refusal.	Guard intent: after refusal, only direct PTP offer is allowed.
High latency	VAD silence too long, TTS slow, or LLM retries.	Log STT/LLM/TTS separately; lower trailing silence and response length.
Wrong language switch	STT returned short confirmation in another script.	Preserve current conversation language for short yes/no fragments.

12. Production readiness checklist

- Agent server is on HTTPS and reachable from VICIdial/Asterisk.
- Gemini, STT, and TTS keys are configured and not committed to git.
- VICIdial API user is dedicated, scoped, and tested.
- Lead data mapping is verified for first_name, phone_number, loan_id, due_amount, lead_id, and campaign.

- Telephony bridge tested with internal extension and real RTP audio.
- PTP creation updates your database and returns a final disposition to VICIdial.
- Per-turn logs include STT, LLM, TTS, and total latency.
- Call recordings/transcripts are stored according to retention and privacy policy.
- Fallback behavior is defined for STT failure, LLM failure, TTS failure, and API timeout.
- Compliance review completed for recovery language, consent, CIBIL/fees wording, and escalation rules.

13. Reference documents

- VICIdial official overview and architecture notes: <https://www.vicidial.org/vicidial.php>
- VICIdial Non-Agent API document: https://vicidial.org/docs/NON-AGENT_API.txt
- VICIdial Agent API document: https://www.vicidial.org/docs/AGENT_API.txt
- VICIdial Call URL features document: https://github.com/inktel/Vicidial/blob/master/docs/CALL_URL_FEATURES.txt
- Asterisk External IVR Interface documentation: <https://docs.asterisk.org/Configuration/Applications/External-IVR-Interface/>

Note: Always verify feature names and available variables on your exact VICIdial build. VICIdial installations vary by version, customization, user-group permissions, and Asterisk module availability.